

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29.8	9.9	1.1	0	0	59.2
U3 - Toxicity prior Beta(0.15, 0.85)	30.5	8.5	0.8	0	0	60.2
U2 - Toxicity prior Beta(0.3, 0.7)	30	11	1.2	0.2	0	57.6
U3 - Toxicity prior Beta(0.3, 0.7)	30.3	10.6	1.2	0	0	57.9
U2 - Toxicity prior Beta(0.5, 0.5)	31.3	13.6	2.7	0.3	0	52.1
U3 - Toxicity prior Beta(0.5, 0.5)	32.2	13.1	1.2	0	0	53.5
U2 - Toxicity prior Beta(1, 1)	31.7	12.9	2	0.3	0	53.1
U3 - Toxicity prior Beta(1, 1)	31.7	12.7	1.8	0	0	53.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.5	5.7	1.4	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	15.8	5.4	1.4	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	15.5	5.9	1.5	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	15.8	5.6	1.5	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	14.9	6.3	2.2	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	15.5	5.9	2.1	0.8	0.2	
U2 - Toxicity prior Beta(1, 1)	15	6.3	2.2	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	15.4	5.9	2.2	0.8	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	3.2	0.5	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1	3.1	0.5	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1	3.3	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.1	0.5	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	3.2	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	3.1	0.6	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	1.2	3.2	0.6	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	1.2	3.1	0.6	0.4	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29	24.8	36.2	8.5	0.8	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	31.7	25.5	34.5	7	0.5	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	25.8	25.7	38.2	8.8	0.8	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	29.4	26	36	7.4	0.5	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	21.5	23.3	41.6	11.9	1.1	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	27.3	23.4	37.8	10	0.9	0.6
U2 - Toxicity prior Beta(1, 1)	20.3	24.2	41.6	12.1	1.1	0.7
U3 - Toxicity prior Beta(1, 1)	25.8	24.1	39.2	9.5	0.8	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.4	8.5	8.2	2.5	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	10.9	8.7	7.7	2.3	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	10	8.5	8.4	2.7	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	8.7	7.8	2.5	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	8.7	8.7	9	3.1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	9.4	8.8	8.6	2.9	0.3	
U2 - Toxicity prior Beta(1, 1)	8.4	8.7	9.2	3.2	0.4	
U3 - Toxicity prior Beta(1, 1)	9.1	8.8	8.8	2.9	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.7	1.7	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.7	1.6	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.7	1.8	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.7	1.7	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.4	2	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.4	1.8	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.6	3.4	2	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.4	1.9	0.7	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.8	51.6	25.3	7.5	0.8	3
U3 - Toxicity prior Beta(0.15, 0.85)	13.1	57.2	21.2	5	0.6	2.9
U2 - Toxicity prior Beta(0.3, 0.7)	10.8	51.9	26.3	7.3	1.1	2.6
U3 - Toxicity prior Beta(0.3, 0.7)	12.3	57.1	21.6	5.7	0.7	2.6
U2 - Toxicity prior Beta(0.5, 0.5)	9.3	49	28.7	9.5	1.2	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	11.5	54.6	23.4	7.4	0.7	2.4
U2 - Toxicity prior Beta(1, 1)	9.3	48.8	28.6	9.4	1.6	2.3
U3 - Toxicity prior Beta(1, 1)	10.5	54.7	24.5	7	0.9	2.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8	12.4	6.4	2.5	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	8.2	13.2	5.8	2.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	7.6	12.3	6.7	2.6	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	7.9	13.1	6	2.3	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	6.4	12.7	7.2	3	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	6.6	13.5	6.5	2.7	0.4	
U2 - Toxicity prior Beta(1, 1)	6.4	12.6	7.2	3	0.6	
U3 - Toxicity prior Beta(1, 1)	6.6	13.5	6.6	2.6	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	4.3	1.8	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	4.3	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	4.2	1.8	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	4.3	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.7	1.9	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.8	1.7	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	3.7	1.9	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.5	3.8	1.7	0.8	0.1	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	6.4	8.4	21.1	53.3	10.6	0.2
U3 - Toxicity prior Beta(0.15, 0.85)	9.7	9.9	20.3	51.6	8.3	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	6.8	8.1	18.7	53.4	12.8	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	9	9.3	18.7	53.3	9.5	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	5.2	7.1	14.7	58.6	14.1	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	8.7	15.1	56.3	11.4	0.2
U2 - Toxicity prior Beta(1, 1)	5.5	7.2	15.6	56.8	14.7	0.2
U3 - Toxicity prior Beta(1, 1)	8	9.6	14.7	56.3	11.2	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.5	5.1	7.4	10.4	2.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4.8	5.1	7.4	10.4	2.3	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	4.9	7.2	10.6	3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	5	7.1	10.8	2.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.9	4.7	6.8	11.3	3.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.2	4.8	6.8	11.4	2.8	
U2 - Toxicity prior Beta(1, 1)	3.9	4.6	6.9	11.3	3.3	
U3 - Toxicity prior Beta(1, 1)	4.1	4.8	6.9	11.4	2.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.7	2.5	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.7	2.4	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3	1.7	2.4	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3.1	1.7	2.4	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.6	2.4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.6	2.3	0.6	
U2 - Toxicity prior Beta(1, 1)	0.2	2.8	1.6	2.4	0.7	
U3 - Toxicity prior Beta(1, 1)	0.2	2.9	1.6	2.3	0.6	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	4.5	35.9	26.5	23.3	9.4	0.4
U3 - Toxicity prior Beta(0.15, 0.85)	7.4	43.1	25.2	17.9	6	0.4
U2 - Toxicity prior Beta(0.3, 0.7)	4.8	35.2	26.5	22.8	10.3	0.4
U3 - Toxicity prior Beta(0.3, 0.7)	7.4	41	26	18.8	6.4	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	5.3	31.9	26.7	24.1	11.6	0.4
U3 - Toxicity prior Beta(0.5, 0.5)	6.9	39.9	26.2	19	7.6	0.4
U2 - Toxicity prior Beta(1, 1)	5.1	32.5	26.1	24.1	11.8	0.4
U3 - Toxicity prior Beta(1, 1)	6.7	40	27.5	18.3	7.1	0.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	8.1	7.3	7	3.5	
U3 - Toxicity prior Beta(0.15, 0.85)	4.3	8.8	7.4	6.5	2.8	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	7.7	7.2	7.2	3.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.2	8.5	7.5	6.7	3	
U2 - Toxicity prior Beta(0.5, 0.5)	3.7	7.5	7.2	7.4	4.1	
U3 - Toxicity prior Beta(0.5, 0.5)	3.9	8.3	7.4	7.1	3.3	
U2 - Toxicity prior Beta(1, 1)	3.7	7.5	7.2	7.4	4.2	
U3 - Toxicity prior Beta(1, 1)	3.8	8.2	7.5	7.1	3.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.3	2	2.1	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.4	2.1	2	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	3.2	2	2.2	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.3	2.1	2	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	3	2	2.3	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.1	2.1	2.1	0.9	
U2 - Toxicity prior Beta(1, 1)	0.1	3	2.1	2.3	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	3.1	2.1	2.1	0.9	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	65.4	16.4	1.9	1	0.1	15.2
U3 - Toxicity prior Beta(0.15, 0.85)	67.1	15.7	1.4	0.7	0.1	15
U2 - Toxicity prior Beta(0.3, 0.7)	66.1	16.5	2.3	0.8	0.1	14.2
U3 - Toxicity prior Beta(0.3, 0.7)	67.1	15.5	2	0.9	0	14.5
U2 - Toxicity prior Beta(0.5, 0.5)	63.5	20.2	2.8	0.8	0.1	12.6
U3 - Toxicity prior Beta(0.5, 0.5)	64.5	20	2	0.6	0.2	12.7
U2 - Toxicity prior Beta(1, 1)	63.8	19.8	3	0.7	0.1	12.6
U3 - Toxicity prior Beta(1, 1)	64.4	19.6	2.1	0.6	0.2	13.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19	6.2	1.6	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	19.4	6.1	1.3	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	18.9	6.3	1.7	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	19.3	6.2	1.4	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	17.8	7	2.1	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	18.3	6.8	1.9	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	17.8	7	2.2	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	18.2	6.8	1.9	0.6	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.7	0.5	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.6	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.7	0.5	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.6	0.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.7	0.6	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.6	0.5	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.8	2.6	0.7	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.8	2.6	0.6	0.2	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.2	55.8	22.4	6.8	0.5	1.3
U3 - Toxicity prior Beta(0.15, 0.85)	15.3	59.6	19.6	4.1	0.3	1.1
U2 - Toxicity prior Beta(0.3, 0.7)	12.1	55.8	23.8	6.6	0.7	1
U3 - Toxicity prior Beta(0.3, 0.7)	14	60.4	20	4.4	0.3	0.9
U2 - Toxicity prior Beta(0.5, 0.5)	9.8	54.3	26.1	8.3	0.7	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	12.3	59.2	21.9	5.4	0.4	0.8
U2 - Toxicity prior Beta(1, 1)	9.9	54.7	25.9	7.9	0.7	0.9
U3 - Toxicity prior Beta(1, 1)	11.7	60.3	21.3	5.5	0.4	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.6	14.5	5.1	1.4	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9	15.1	4.5	1.1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.5	14.4	5.3	1.5	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	8.9	15.2	4.6	1.1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	7.2	14.9	5.9	1.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	7.7	15.7	5	1.3	0.2	
U2 - Toxicity prior Beta(1, 1)	7.2	14.8	6	1.6	0.2	
U3 - Toxicity prior Beta(1, 1)	7.6	15.8	5	1.3	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.9	1.3	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.9	1	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.9	1.3	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.9	1.1	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3.5	1.4	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	1.1	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.3	3.4	1.4	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	3.4	1.1	0.3	0.1	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	44.6	9.5	0.9	0.1	0.1	44.8
U3 - Toxicity prior Beta(0.15, 0.85)	46.6	8	0.9	0.1	0	44.4
U2 - Toxicity prior Beta(0.3, 0.7)	45.4	10.2	1.2	0.5	0.1	42.6
U3 - Toxicity prior Beta(0.3, 0.7)	46.3	9.5	0.9	0.2	0	43.1
U2 - Toxicity prior Beta(0.5, 0.5)	45.6	13	1.8	0.4	0.1	39.1
U3 - Toxicity prior Beta(0.5, 0.5)	46.8	12.2	1.5	0.4	0	39.1
U2 - Toxicity prior Beta(1, 1)	46.1	12.5	2	0.4	0.1	38.9
U3 - Toxicity prior Beta(1, 1)	48.4	11	1.2	0.3	0	39.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.1	5.4	1.2	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	17.3	5.3	1.2	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	17.1	5.5	1.4	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	17.3	5.4	1.3	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	16.3	6.1	2	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	16.7	5.9	1.9	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	16.2	6.1	2.1	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	16.6	5.9	2	0.6	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	2.7	0.4	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	2.6	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	2.7	0.5	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	2.7	0.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1	2.7	0.6	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1	2.7	0.6	0.2	0	
U2 - Toxicity prior Beta(1, 1)	1	2.8	0.6	0.3	0	
U3 - Toxicity prior Beta(1, 1)	1	2.7	0.6	0.2	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.9	20.1	23.3	14.5	16.2	7
U3 - Toxicity prior Beta(0.15, 0.85)	25.9	22.3	18.6	12.8	13.6	6.8
U2 - Toxicity prior Beta(0.3, 0.7)	17.5	19.8	23.7	14.6	18.3	6.1
U3 - Toxicity prior Beta(0.3, 0.7)	24.1	20.7	20.2	13.5	15.1	6.4
U2 - Toxicity prior Beta(0.5, 0.5)	14.5	18.1	25.1	15.3	23.2	3.8
U3 - Toxicity prior Beta(0.5, 0.5)	22.7	20.2	20.5	13.8	18.9	3.9
U2 - Toxicity prior Beta(1, 1)	15.7	17.8	24.5	15.8	22.6	3.6
U3 - Toxicity prior Beta(1, 1)	23.8	19.6	20.6	13.8	18.5	3.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.3	7	5.9	4.9	3.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9	6.8	5.8	4.8	3	
U2 - Toxicity prior Beta(0.3, 0.7)	8.1	6.8	5.9	5	3.6	
U3 - Toxicity prior Beta(0.3, 0.7)	8.7	6.6	5.8	4.9	3.4	
U2 - Toxicity prior Beta(0.5, 0.5)	7	6.5	6.6	5.3	4.2	
U3 - Toxicity prior Beta(0.5, 0.5)	7.6	6.5	6.3	5.3	3.9	
U2 - Toxicity prior Beta(1, 1)	6.9	6.5	6.6	5.3	4.2	
U3 - Toxicity prior Beta(1, 1)	7.6	6.5	6.3	5.2	4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4	1.7	1.6	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.6	1.6	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.9	1.6	1.6	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.9	1.6	1.6	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	1.7	1.8	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	1.6	1.7	0.8	
U2 - Toxicity prior Beta(1, 1)	0.7	3.5	1.7	1.8	0.9	
U3 - Toxicity prior Beta(1, 1)	0.7	3.5	1.6	1.7	0.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.4	38.3	12.4	2.9	0.4	27.6
U3 - Toxicity prior Beta(0.15, 0.85)	19.4	38.3	11.3	2.3	0.3	28.4
U2 - Toxicity prior Beta(0.3, 0.7)	16.3	39.6	13.5	3.6	0.4	26.6
U3 - Toxicity prior Beta(0.3, 0.7)	17.7	40	12.3	2.8	0.3	26.9
U2 - Toxicity prior Beta(0.5, 0.5)	15	43.2	15.7	4.6	0.5	21
U3 - Toxicity prior Beta(0.5, 0.5)	16.6	45.2	13.2	3.7	0.4	20.9
U2 - Toxicity prior Beta(1, 1)	15	43.2	15.9	4.8	0.5	20.6
U3 - Toxicity prior Beta(1, 1)	16.8	44.6	13.4	3.6	0.4	21.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.9	10.2	3.8	1.2	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.9	10.4	3.5	1.1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	11.5	10.3	4	1.4	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	11.7	10.6	3.7	1.2	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	9.8	11.4	4.8	1.6	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	9.9	11.8	4.5	1.5	0.2	
U2 - Toxicity prior Beta(1, 1)	9.8	11.3	4.8	1.6	0.2	
U3 - Toxicity prior Beta(1, 1)	9.9	11.7	4.6	1.5	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4.3	1	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	4.2	0.9	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	4.2	1	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	4.3	0.9	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.8	1.2	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.8	1.1	0.4	0	
U2 - Toxicity prior Beta(1, 1)	0.6	3.7	1.2	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.6	3.7	1.1	0.4	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	68.4	21.5	2.5	0.1	0	7.5
U3 - Toxicity prior Beta(0.15, 0.85)	70.8	20	2	0.1	0	7.1
U2 - Toxicity prior Beta(0.3, 0.7)	66.9	23.8	2.8	0.2	0	6.3
U3 - Toxicity prior Beta(0.3, 0.7)	68.7	22.3	2.6	0.1	0	6.3
U2 - Toxicity prior Beta(0.5, 0.5)	64.2	25.2	5.1	0.2	0	5.3
U3 - Toxicity prior Beta(0.5, 0.5)	66.4	24.3	3.9	0	0	5.4
U2 - Toxicity prior Beta(1, 1)	64.7	25.7	4.5	0.2	0	4.9
U3 - Toxicity prior Beta(1, 1)	66.1	24.8	3.5	0.1	0	5.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.1	7.7	2.4	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	18.7	7.5	2.1	0.7	0	
U2 - Toxicity prior Beta(0.3, 0.7)	17.8	7.9	2.5	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	18.5	7.6	2.2	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	16.6	8.4	3.1	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	17.1	8.2	2.9	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	16.7	8.4	3.1	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	17.1	8.2	3	0.9	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.7	0.8	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.6	0.7	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.7	0.8	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.6	0.7	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	3.6	0.9	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	3.5	0.9	0.4	0	
U2 - Toxicity prior Beta(1, 1)	1.2	3.5	0.9	0.4	0	
U3 - Toxicity prior Beta(1, 1)	1.2	3.5	0.9	0.4	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.7	16.3	11.4	44.7	15.6	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	16.9	16.2	10.5	41.7	14.4	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	11	15.7	11.5	43.8	17.7	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	17.5	15.9	9.8	40.6	15.9	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	9.5	13.3	9.7	46.1	21.2	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	15.5	15.5	8.4	42.5	17.8	0.3
U2 - Toxicity prior Beta(1, 1)	10.2	13	10.7	45.5	20.3	0.3
U3 - Toxicity prior Beta(1, 1)	15.6	14.6	8.7	42.4	18.3	0.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.9	5.8	6.1	9.1	4.1	
U3 - Toxicity prior Beta(0.15, 0.85)	5.4	5.9	5.9	9	3.8	
U2 - Toxicity prior Beta(0.3, 0.7)	4.7	5.6	5.8	9.4	4.4	
U3 - Toxicity prior Beta(0.3, 0.7)	5.1	5.6	5.7	9.4	4.1	
U2 - Toxicity prior Beta(0.5, 0.5)	4.4	5.2	5.6	9.9	4.9	
U3 - Toxicity prior Beta(0.5, 0.5)	4.7	5.4	5.5	9.9	4.5	
U2 - Toxicity prior Beta(1, 1)	4.3	5.2	5.6	10	4.9	
U3 - Toxicity prior Beta(1, 1)	4.7	5.3	5.5	10	4.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.3	1.7	2.4	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.3	1.6	2.3	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.2	1.7	2.3	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.2	1.6	2.3	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.6	2.3	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.1	1.6	2.1	1	
U2 - Toxicity prior Beta(1, 1)	0.3	3	1.6	2.2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.4	3.1	1.6	2.2	1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.3	40.3	11.4	3.5	0.5	24
U3 - Toxicity prior Beta(0.15, 0.85)	21.9	39.7	10.6	2.5	0.4	24.9
U2 - Toxicity prior Beta(0.3, 0.7)	18.7	41.2	12.9	3.9	0.7	22.6
U3 - Toxicity prior Beta(0.3, 0.7)	20.9	40.3	13.2	2.5	0.5	22.6
U2 - Toxicity prior Beta(0.5, 0.5)	17.4	42.4	14.8	5.5	0.8	19.1
U3 - Toxicity prior Beta(0.5, 0.5)	19.8	44	13.6	3.4	0.8	18.4
U2 - Toxicity prior Beta(1, 1)	16.9	44.5	14	5.3	0.8	18.5
U3 - Toxicity prior Beta(1, 1)	19.6	43.8	13.2	4.1	0.7	18.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12	9.6	4.1	1.7	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	12.2	9.8	3.8	1.6	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	11.5	9.9	4.2	1.9	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	11.7	10.1	4	1.7	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	9.8	10.5	5.2	2.3	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	10.2	10.7	4.9	2.1	0.4	
U2 - Toxicity prior Beta(1, 1)	9.8	10.6	5.2	2.3	0.5	
U3 - Toxicity prior Beta(1, 1)	10.1	10.8	4.9	2.2	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.5	1.1	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4.5	1	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4.5	1.1	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	4.5	1.1	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	4	1.3	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	4	1.2	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	4	1.3	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.7	4	1.2	0.7	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.6	21.2	19.6	28.5	9.6	1.5
U3 - Toxicity prior Beta(0.15, 0.85)	27.6	24.1	14.1	25.4	7.5	1.3
U2 - Toxicity prior Beta(0.3, 0.7)	17.8	20.7	18.8	30.5	11	1.2
U3 - Toxicity prior Beta(0.3, 0.7)	26	24	14	25.6	9.2	1.2
U2 - Toxicity prior Beta(0.5, 0.5)	15.9	20	18.7	32	12.5	0.9
U3 - Toxicity prior Beta(0.5, 0.5)	23.6	23.9	12.4	29.4	9.7	1
U2 - Toxicity prior Beta(1, 1)	15.1	21.4	17.7	31.4	13.4	1
U3 - Toxicity prior Beta(1, 1)	24.3	23.3	12.7	28.6	10.2	0.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.7	7	6.3	6.8	3.1	
U3 - Toxicity prior Beta(0.15, 0.85)	7.3	7	5.9	6.7	2.9	
U2 - Toxicity prior Beta(0.3, 0.7)	6.4	6.8	6.1	7.1	3.5	
U3 - Toxicity prior Beta(0.3, 0.7)	7.1	6.7	5.8	7	3.2	
U2 - Toxicity prior Beta(0.5, 0.5)	5.9	6.3	6.1	7.7	4	
U3 - Toxicity prior Beta(0.5, 0.5)	6.6	6.3	5.8	7.6	3.6	
U2 - Toxicity prior Beta(1, 1)	5.8	6.4	6	7.7	4	
U3 - Toxicity prior Beta(1, 1)	6.5	6.4	5.7	7.6	3.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.8	1.8	2.1	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.7	2	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.8	1.8	2.1	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.7	1.7	2.1	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.5	1.7	2.1	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	1.7	2.1	0.8	
U2 - Toxicity prior Beta(1, 1)	0.6	3.5	1.7	2.1	0.9	
U3 - Toxicity prior Beta(1, 1)	0.7	3.5	1.7	2.1	0.8	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	46.8	35.2	12.3	3	0.1	2.6
U3 - Toxicity prior Beta(0.15, 0.85)	54.3	32.3	8.5	2.2	0.1	2.6
U2 - Toxicity prior Beta(0.3, 0.7)	45.9	35.7	12.9	3.5	0.3	1.7
U3 - Toxicity prior Beta(0.3, 0.7)	54.3	33	8.1	2.3	0.1	2.2
U2 - Toxicity prior Beta(0.5, 0.5)	45.2	36.3	12.5	4.2	0.7	1.1
U3 - Toxicity prior Beta(0.5, 0.5)	54.1	33.9	7.7	2.7	0.3	1.3
U2 - Toxicity prior Beta(1, 1)	46.4	34.2	13	4.6	0.5	1.3
U3 - Toxicity prior Beta(1, 1)	54	34.5	7.4	2.6	0.3	1.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.2	9.2	5.4	3.1	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	12.1	8.8	5.1	3	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	10.7	9.1	5.6	3.4	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	11.7	8.6	5.2	3.3	1	
U2 - Toxicity prior Beta(0.5, 0.5)	9.9	9	6	3.7	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	10.9	8.7	5.5	3.5	1.2	
U2 - Toxicity prior Beta(1, 1)	9.8	9	6.1	3.7	1.3	
U3 - Toxicity prior Beta(1, 1)	10.8	8.7	5.6	3.6	1.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	4.4	1.8	1.3	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	4.2	1.7	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	4.4	1.8	1.3	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	4.2	1.7	1.3	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	4.2	1.8	1.4	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	1.3	4	1.7	1.3	0.4	
U2 - Toxicity prior Beta(1, 1)	1.2	4.1	1.8	1.4	0.4	
U3 - Toxicity prior Beta(1, 1)	1.3	4	1.7	1.4	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	36	27.9	14.9	13.9	7.3	0
U3 - Toxicity prior Beta(0.15, 0.85)	41.2	30.4	14.1	10	4.3	0
U2 - Toxicity prior Beta(0.3, 0.7)	36.4	27.9	15.4	13.2	7.1	0
U3 - Toxicity prior Beta(0.3, 0.7)	41.9	30.2	13.7	9.6	4.6	0
U2 - Toxicity prior Beta(0.5, 0.5)	35.6	27	16.1	13.3	8	0
U3 - Toxicity prior Beta(0.5, 0.5)	41.5	29.2	15	9.8	4.5	0
U2 - Toxicity prior Beta(1, 1)	36.2	27.3	16	13	7.5	0
U3 - Toxicity prior Beta(1, 1)	40.9	30.1	14.3	10.1	4.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.3	5.3	6.1	8.2	6.1	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	5.7	6.2	8	5.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.2	5.2	6.1	8.2	6.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	5.7	6.2	8	5.6	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	5.1	6	8.3	6.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	5.5	6.2	8.1	5.7	
U2 - Toxicity prior Beta(1, 1)	4.1	5.1	5.9	8.3	6.4	
U3 - Toxicity prior Beta(1, 1)	4.3	5.5	6.2	8.2	5.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	2.2	2.9	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.9	2.2	2.9	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	2.2	2.9	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	2.2	2.8	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	2.1	3	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	2.2	2.9	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	2.1	3	1.5	
U3 - Toxicity prior Beta(1, 1)	0.2	2.8	2.3	2.9	1.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.7	66.5	14	0.7	0	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	21.6	67.9	10.1	0.3	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	16.8	66.8	15	1.3	0	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	20.2	67.2	11.7	0.7	0	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	15.9	65.5	17.1	1.4	0	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	20.5	65.7	12.9	0.8	0	0.1
U2 - Toxicity prior Beta(1, 1)	15.3	66.6	16.8	1.2	0	0.1
U3 - Toxicity prior Beta(1, 1)	19.7	67.3	12	0.8	0	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.6	15.9	4.1	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	10.3	16.2	3.2	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	9.4	15.9	4.1	0.5	0	
U3 - Toxicity prior Beta(0.3, 0.7)	10	16.2	3.4	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	8.6	16.2	4.7	0.5	0	
U3 - Toxicity prior Beta(0.5, 0.5)	9.2	16.7	3.7	0.4	0	
U2 - Toxicity prior Beta(1, 1)	8.4	16.3	4.7	0.6	0	
U3 - Toxicity prior Beta(1, 1)	9.2	16.6	3.7	0.4	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	4.1	1.2	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4	0.8	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	4	1.2	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	4	0.9	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.7	1.3	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.7	0.9	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.5	3.7	1.3	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.6	3.7	0.9	0.2	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.7	11.6	10.6	18.4	49.7	0
U3 - Toxicity prior Beta(0.15, 0.85)	16.1	15.2	9.5	16.1	43.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	9.7	11.3	9.3	17	52.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	15	14.5	9.9	14.7	45.9	0
U2 - Toxicity prior Beta(0.5, 0.5)	8.5	8.8	10	16.6	56.1	0
U3 - Toxicity prior Beta(0.5, 0.5)	14.6	12.9	9.3	14.1	49.1	0
U2 - Toxicity prior Beta(1, 1)	8	9	9.1	16.3	57.6	0
U3 - Toxicity prior Beta(1, 1)	14.6	12.9	8.9	14	49.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.4	4.1	5.2	7.3	9.9	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	4.2	5.4	7.5	9.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	4	5.2	7.3	10.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	4.1	5.2	7.4	9.8	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.8	5	7.3	10.7	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	3.9	5.2	7.4	10.1	
U2 - Toxicity prior Beta(1, 1)	3.3	3.8	4.9	7.1	10.9	
U3 - Toxicity prior Beta(1, 1)	3.4	3.9	5.1	7.4	10.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.5	1.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.6	2.5	1.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.5	2.5	1.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.5	1.7	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.5	1.8	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.5	2.5	1.7	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.2	60	16.7	0.7	0	4.4
U3 - Toxicity prior Beta(0.15, 0.85)	22.2	58.8	14.1	0.5	0	4.4
U2 - Toxicity prior Beta(0.3, 0.7)	16	62.6	17.3	0.8	0	3.3
U3 - Toxicity prior Beta(0.3, 0.7)	18.9	60.6	16.1	0.9	0	3.5
U2 - Toxicity prior Beta(0.5, 0.5)	14.8	63.1	18.4	1.1	0	2.6
U3 - Toxicity prior Beta(0.5, 0.5)	18	63.1	15.6	0.7	0	2.6
U2 - Toxicity prior Beta(1, 1)	14.3	63.6	18.7	1	0	2.4
U3 - Toxicity prior Beta(1, 1)	15.9	63.5	16.6	1	0	3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.3	13.2	5.6	1.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	13.4	5.3	1.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.7	13.3	6	1.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	9.2	13.5	5.5	1.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	7.7	13.7	6.4	1.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.1	13.8	6.1	1.7	0.2	
U2 - Toxicity prior Beta(1, 1)	7.7	13.6	6.5	1.9	0.2	
U3 - Toxicity prior Beta(1, 1)	8	13.9	6.1	1.7	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.9	1.7	0.7	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.9	1.5	0.6	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4.9	1.7	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	4.8	1.6	0.6	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	4.5	1.8	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	4.4	1.7	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	4.4	1.7	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.8	4.4	1.6	0.7	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	50.5	30.2	8.9	0.7	0.5
U3 - Toxicity prior Beta(0.15, 0.85)	13.7	53.8	24.1	7.5	0.4	0.5
U2 - Toxicity prior Beta(0.3, 0.7)	10.2	46.4	31.9	10	1	0.5
U3 - Toxicity prior Beta(0.3, 0.7)	13.1	50	27.9	7.9	0.6	0.5
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	46.1	31	10.5	1.4	0.5
U3 - Toxicity prior Beta(0.5, 0.5)	14.7	48.4	27.4	8.5	0.6	0.4
U2 - Toxicity prior Beta(1, 1)	10.1	45.6	32.3	10.3	1.1	0.6
U3 - Toxicity prior Beta(1, 1)	14.4	48.4	27.2	8.9	0.7	0.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.2	11	8.4	4.2	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	5.6	11.4	7.9	4	1	
U2 - Toxicity prior Beta(0.3, 0.7)	5.1	10.3	8.6	4.7	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	5.4	10.9	8.2	4.3	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.5	10.1	8.6	5.2	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	10.6	8.2	4.8	1.4	
U2 - Toxicity prior Beta(1, 1)	4.5	10.1	8.6	5.2	1.5	
U3 - Toxicity prior Beta(1, 1)	5	10.5	8.3	4.8	1.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	4.1	2.4	1.5	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	4	2.3	1.5	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.9	2.5	1.6	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.9	2.4	1.5	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.6	2.4	1.7	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.7	2.3	1.6	0.4	
U2 - Toxicity prior Beta(1, 1)	0.4	3.6	2.4	1.7	0.5	
U3 - Toxicity prior Beta(1, 1)	0.5	3.7	2.3	1.6	0.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	54	1.2	0	0	0	44.8
U3 - Toxicity prior Beta(0.15, 0.85)	54.4	0.8	0	0	0	44.8
U2 - Toxicity prior Beta(0.3, 0.7)	55.4	1	0	0	0	43.6
U3 - Toxicity prior Beta(0.3, 0.7)	54.8	1.1	0	0	0	44.1
U2 - Toxicity prior Beta(0.5, 0.5)	55.7	1.5	0	0	0	42.8
U3 - Toxicity prior Beta(0.5, 0.5)	55.4	1.1	0	0	0	43.5
U2 - Toxicity prior Beta(1, 1)	55	1.4	0	0	0	43.6
U3 - Toxicity prior Beta(1, 1)	55.6	1	0	0	0	43.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.9	2.3	0	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20	2.2	0	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	20.1	2.3	0	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	20.1	2.2	0	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	20	2.5	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	20	2.4	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	19.9	2.5	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	20	2.4	0.1	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U2 - Toxicity prior Beta(1, 1)	0.2	1.1	0	0	0	
U3 - Toxicity prior Beta(1, 1)	0.3	1.1	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.6	64.9	10	1.4	0	2.1
U3 - Toxicity prior Beta(0.15, 0.85)	25.3	61.1	9.6	1.7	0	2.3
U2 - Toxicity prior Beta(0.3, 0.7)	20.5	64.2	12.3	1.6	0	1.4
U3 - Toxicity prior Beta(0.3, 0.7)	23.1	62.5	11.3	1.6	0	1.5
U2 - Toxicity prior Beta(0.5, 0.5)	20	62	15.1	1.4	0	1.5
U3 - Toxicity prior Beta(0.5, 0.5)	22.9	60.5	13.9	1.3	0	1.4
U2 - Toxicity prior Beta(1, 1)	19.2	62.2	15.4	1.6	0	1.6
U3 - Toxicity prior Beta(1, 1)	23.3	60.2	13.8	1.1	0	1.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.4	13.9	5.1	1.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	13.8	4.8	1.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.9	13.7	5.4	1.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.4	13.7	5.1	1.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	13.7	5.9	1.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	13.7	5.5	1.8	0.1	
U2 - Toxicity prior Beta(1, 1)	8.3	13.7	5.8	1.9	0.1	
U3 - Toxicity prior Beta(1, 1)	8.8	13.7	5.5	1.8	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	5	1.6	0.6	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	4.9	1.6	0.6	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1	4.9	1.7	0.6	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1	4.9	1.6	0.6	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1	4.7	1.8	0.6	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	4.5	1.7	0.6	0	
U2 - Toxicity prior Beta(1, 1)	1	4.7	1.8	0.6	0	
U3 - Toxicity prior Beta(1, 1)	1.1	4.5	1.7	0.6	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.5	2	0	0	0	80.5
U3 - Toxicity prior Beta(0.15, 0.85)	17.7	1.7	0	0	0	80.6
U2 - Toxicity prior Beta(0.3, 0.7)	18	2.2	0	0	0	79.8
U3 - Toxicity prior Beta(0.3, 0.7)	18.3	2.1	0	0	0	79.6
U2 - Toxicity prior Beta(0.5, 0.5)	18.2	2.1	0	0	0	79.7
U3 - Toxicity prior Beta(0.5, 0.5)	17.5	2.6	0	0	0	79.9
U2 - Toxicity prior Beta(1, 1)	18.2	1.8	0	0	0	80
U3 - Toxicity prior Beta(1, 1)	17.8	2.6	0	0	0	79.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.4	4	0.3	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	15.5	3.9	0.3	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	15.5	4.1	0.4	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	15.6	4	0.3	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	15.5	4.2	0.7	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	15.5	4.2	0.7	0	0	
U2 - Toxicity prior Beta(1, 1)	15.5	4.2	0.8	0	0	
U3 - Toxicity prior Beta(1, 1)	15.5	4.2	0.8	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	2.4	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	2.3	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	2.4	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2.4	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.3	0.2	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.3	0.2	0	0	
U2 - Toxicity prior Beta(1, 1)	0.7	2.3	0.2	0	0	
U3 - Toxicity prior Beta(1, 1)	0.7	2.2	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	1.3	0.8	3	53.4	41.5	0
U3 - Toxicity prior Beta(0.15, 0.85)	2.3	2	3.7	57.2	34.8	0
U2 - Toxicity prior Beta(0.3, 0.7)	1.3	0.7	2.8	53.5	41.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	2.4	2.1	3.2	57.1	35.2	0
U2 - Toxicity prior Beta(0.5, 0.5)	1	1	3.1	52.8	42.1	0
U3 - Toxicity prior Beta(0.5, 0.5)	2	2.3	3.9	56.5	35.3	0
U2 - Toxicity prior Beta(1, 1)	1.2	1.1	3.1	53	41.6	0
U3 - Toxicity prior Beta(1, 1)	2.3	2.4	3.3	56.3	35.7	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	3.8	12.7	7.2	
U3 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	4	13.4	6.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.1	3.2	3.8	12.8	7.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.1	3.2	3.9	13.4	6.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	3.8	12.8	7.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	3.9	13.4	6.5	
U2 - Toxicity prior Beta(1, 1)	3	3.1	3.8	12.8	7.2	
U3 - Toxicity prior Beta(1, 1)	3.1	3.1	3.9	13.4	6.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1	2.4	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1.1	2.4	1.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1	2.4	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1.1	2.4	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.4	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.4	1.3	
U2 - Toxicity prior Beta(1, 1)	0	2.4	1	2.4	1.5	
U3 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.4	1.3	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.6	37.4	32.6	3.3	0	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	33.1	35.6	28.5	2.7	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	25.7	35.4	34.9	3.9	0	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	31.5	33.8	31.4	3.2	0	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	24.4	34.3	37.1	4	0	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	30.4	33.1	32.9	3.5	0	0.1
U2 - Toxicity prior Beta(1, 1)	25	33.3	37.4	4.2	0	0.1
U3 - Toxicity prior Beta(1, 1)	31.1	33.3	32.1	3.4	0	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.1	10.5	8.2	2	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.7	10.6	7.8	1.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	10.4	8.5	2.2	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	9.4	10.4	8.1	1.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	7.9	10.2	9.2	2.4	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.6	10.4	8.7	2.2	0.2	
U2 - Toxicity prior Beta(1, 1)	8	10.3	9.1	2.4	0.2	
U3 - Toxicity prior Beta(1, 1)	8.6	10.4	8.7	2.1	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4	2.1	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4	1.9	0.8	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.9	2.1	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.9	2	0.8	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.7	2.2	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.7	2.1	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.7	2.2	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	0.8	3.7	2.1	0.8	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.2	40.6	27.3	3	0	3.9
U3 - Toxicity prior Beta(0.15, 0.85)	30.4	39.2	23.9	2.6	0	3.9
U2 - Toxicity prior Beta(0.3, 0.7)	23.7	39.4	30.7	3	0	3.2
U3 - Toxicity prior Beta(0.3, 0.7)	28.9	38.7	26.3	2.6	0	3.5
U2 - Toxicity prior Beta(0.5, 0.5)	21.3	37.8	34.5	3.5	0.1	2.8
U3 - Toxicity prior Beta(0.5, 0.5)	26	37.7	29.6	3.6	0	3.1
U2 - Toxicity prior Beta(1, 1)	22.4	37.6	33	3.9	0.1	3
U3 - Toxicity prior Beta(1, 1)	25.3	38.1	30.1	3.5	0	3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.1	10.1	7.1	2.2	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	10.6	10	6.8	2	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	9.6	10	7.6	2.4	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	10.3	9.8	7.2	2.2	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	8.4	10.1	8.4	2.6	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	9	9.9	8	2.5	0.2	
U2 - Toxicity prior Beta(1, 1)	8.3	10	8.5	2.7	0.3	
U3 - Toxicity prior Beta(1, 1)	8.9	10	8.1	2.5	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4.4	1.6	1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.3	1.6	0.9	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4.4	1.7	1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	4.2	1.6	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	4	1.8	1.1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	3.9	1.7	1	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	3.9	1.9	1.1	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	3.9	1.8	1	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.1	30.6	43.4	7.6	0	1.3
U3 - Toxicity prior Beta(0.15, 0.85)	21	32.5	40	5.2	0.1	1.2
U2 - Toxicity prior Beta(0.3, 0.7)	15.2	28.7	46.4	8.6	0.1	1
U3 - Toxicity prior Beta(0.3, 0.7)	19.3	30.6	41.4	7.5	0.3	0.9
U2 - Toxicity prior Beta(0.5, 0.5)	14	25.4	48.8	10.7	0.4	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	16.7	28	45.6	8.6	0.4	0.7
U2 - Toxicity prior Beta(1, 1)	14.6	25.6	48.5	10.3	0.4	0.6
U3 - Toxicity prior Beta(1, 1)	17.4	28.1	44.4	9	0.4	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.3	9.3	9.7	3.2	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	7.8	9.3	9.4	3.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	6.9	9	10	3.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	7.4	9.1	9.7	3.3	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	6.1	8.7	10.7	3.9	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	6.7	8.6	10.4	3.7	0.5	
U2 - Toxicity prior Beta(1, 1)	6.1	8.7	10.7	4	0.6	
U3 - Toxicity prior Beta(1, 1)	6.5	8.7	10.4	3.8	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4	2.1	1.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	2.1	1.2	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.9	2.2	1.3	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.9	2.1	1.2	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	2.3	1.3	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.5	2.2	1.2	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	3.5	2.2	1.3	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	3.5	2.1	1.3	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.4	43.1	4.3	0.1	0	25.1
U3 - Toxicity prior Beta(0.15, 0.85)	28.4	42.5	3.3	0	0	25.8
U2 - Toxicity prior Beta(0.3, 0.7)	25.1	46.8	4.9	0.1	0	23.1
U3 - Toxicity prior Beta(0.3, 0.7)	25.4	45.7	4.8	0	0	24.1
U2 - Toxicity prior Beta(0.5, 0.5)	22.9	51.2	5.2	0.2	0	20.5
U3 - Toxicity prior Beta(0.5, 0.5)	24.1	50.3	4.8	0.1	0	20.7
U2 - Toxicity prior Beta(1, 1)	21.9	51.7	5.8	0.2	0	20.4
U3 - Toxicity prior Beta(1, 1)	24.8	48.5	5.5	0.1	0	21.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	13.1	11.2	2.8	0.6	0	
U3 - Toxicity prior Beta(0.15, 0.85)	13.4	11	2.7	0.6	0	
U2 - Toxicity prior Beta(0.3, 0.7)	12.6	11.5	3	0.7	0	
U3 - Toxicity prior Beta(0.3, 0.7)	13.1	11.4	2.9	0.6	0	
U2 - Toxicity prior Beta(0.5, 0.5)	11.4	12.4	3.8	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	11.7	12.4	3.5	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	11.4	12.4	3.8	0.7	0	
U3 - Toxicity prior Beta(1, 1)	11.7	12.4	3.5	0.7	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	4.8	1	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4.6	0.9	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	4.8	1	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.7	1	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1	4.4	1.1	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1	4.3	1	0.3	0	
U2 - Toxicity prior Beta(1, 1)	1	4.3	1.1	0.3	0	
U3 - Toxicity prior Beta(1, 1)	1	4.3	1	0.3	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	62.7	32	2.6	0.1	0	2.6
U3 - Toxicity prior Beta(0.15, 0.85)	65.7	30.1	1.8	0	0	2.4
U2 - Toxicity prior Beta(0.3, 0.7)	60.7	33.9	2.9	0.1	0	2.4
U3 - Toxicity prior Beta(0.3, 0.7)	64.7	30.6	2.2	0	0	2.5
U2 - Toxicity prior Beta(0.5, 0.5)	56.7	37	4	0.1	0	2.2
U3 - Toxicity prior Beta(0.5, 0.5)	61.8	33	2.9	0	0	2.3
U2 - Toxicity prior Beta(1, 1)	59.3	34.6	3.9	0.1	0	2.1
U3 - Toxicity prior Beta(1, 1)	61.8	32.9	3	0	0	2.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.3	9.2	1.7	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	19.1	9	1.3	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	17.9	9.5	1.9	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	18.8	9.2	1.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	17.1	10.1	2	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	18.1	9.7	1.5	0.2	0	
U2 - Toxicity prior Beta(1, 1)	17.1	10.1	2.1	0.3	0	
U3 - Toxicity prior Beta(1, 1)	17.9	9.8	1.6	0.2	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	3.3	0.8	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	3.1	0.5	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.4	0.8	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	3.2	0.6	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	3.2	0.9	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3	0.6	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.9	3.2	0.9	0.1	0	
U3 - Toxicity prior Beta(1, 1)	1	3.1	0.6	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15.9	14.2	15.5	42.8	11.6	0
U3 - Toxicity prior Beta(0.15, 0.85)	22.8	16.8	14	37.4	9	0
U2 - Toxicity prior Beta(0.3, 0.7)	15.6	12.9	15	43.1	13.4	0
U3 - Toxicity prior Beta(0.3, 0.7)	23.6	16.8	12.4	37	10.2	0
U2 - Toxicity prior Beta(0.5, 0.5)	14.9	13.2	12.2	45.5	14.2	0
U3 - Toxicity prior Beta(0.5, 0.5)	21.2	16.3	9.9	40.1	12.5	0
U2 - Toxicity prior Beta(1, 1)	15.7	12.9	11.5	45.8	14.1	0
U3 - Toxicity prior Beta(1, 1)	21.3	14.9	11.9	40	11.9	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	5.5	6.6	9.9	3.4	
U3 - Toxicity prior Beta(0.15, 0.85)	4.9	5.6	6.6	9.8	3.1	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	5.3	6.4	10.2	3.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.7	5.4	6.4	10.1	3.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	5.1	6.1	10.6	4.1	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	5.2	6.1	10.6	3.7	
U2 - Toxicity prior Beta(1, 1)	4.1	5.1	6.1	10.6	4.1	
U3 - Toxicity prior Beta(1, 1)	4.3	5.2	6.1	10.7	3.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.9	2.3	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3	2	2.3	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.9	2.3	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.9	2.3	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.9	2.3	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.9	2.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.8	1.9	2.3	1.2	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	1.9	2.2	1.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	2.8	40.2	29.2	24.5	3.3	0
U3 - Toxicity prior Beta(0.15, 0.85)	3.9	46.9	28.1	19.6	1.5	0
U2 - Toxicity prior Beta(0.3, 0.7)	2.8	39.5	30.7	24.4	2.6	0
U3 - Toxicity prior Beta(0.3, 0.7)	4.3	46.3	28.6	19.1	1.7	0
U2 - Toxicity prior Beta(0.5, 0.5)	2.7	38.5	31	24.3	3.5	0
U3 - Toxicity prior Beta(0.5, 0.5)	4	44.6	29.4	20.2	1.8	0
U2 - Toxicity prior Beta(1, 1)	2.3	38.6	30.4	24.9	3.8	0
U3 - Toxicity prior Beta(1, 1)	4.2	44.6	29.6	20	1.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.9	13.3	7.7	4.4	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4	14.2	7.5	3.9	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	3.9	13.2	7.7	4.5	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4	14.2	7.4	3.9	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	13.1	7.9	4.5	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	3.9	14	7.5	4.1	0.5	
U2 - Toxicity prior Beta(1, 1)	3.7	13.1	7.9	4.5	0.7	
U3 - Toxicity prior Beta(1, 1)	3.9	14	7.6	4	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.7	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.9	1.6	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.7	1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.6	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.8	1.7	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.6	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0	2.7	1.7	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.6	0.9	0.2	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.4	15.4	31.2	32.2	9.8	0
U3 - Toxicity prior Beta(0.15, 0.85)	16.4	19.1	32	26.5	6	0
U2 - Toxicity prior Beta(0.3, 0.7)	12.1	14.4	32.9	30.5	10.1	0
U3 - Toxicity prior Beta(0.3, 0.7)	17.4	18.1	32.5	26	6	0
U2 - Toxicity prior Beta(0.5, 0.5)	11.7	14.2	32.1	30.7	11.3	0
U3 - Toxicity prior Beta(0.5, 0.5)	17	19.3	31.5	26.2	6	0
U2 - Toxicity prior Beta(1, 1)	12.4	13.7	30	32.7	11.2	0
U3 - Toxicity prior Beta(1, 1)	16.5	17.8	32.4	26.5	6.8	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	4.5	8.6	9	4.4	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	4.7	9.1	8.8	3.7	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	4.4	8.5	9	4.7	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	4.6	9	8.9	4	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	4.2	8.1	9.3	5	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	4.4	8.7	9.2	4.2	
U2 - Toxicity prior Beta(1, 1)	3.3	4.2	8.2	9.4	4.9	
U3 - Toxicity prior Beta(1, 1)	3.5	4.5	8.7	9.2	4.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.9	2.9	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	2.1	2.8	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.9	2.9	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	2.1	2.8	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.9	2.9	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	2	2.8	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.9	2.8	1.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	2	2.8	1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	6.8	6.8	11.4	48.8	26.2	0
U3 - Toxicity prior Beta(0.15, 0.85)	12.2	7.3	12.2	46.2	22.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	7.2	5.2	11.9	46.7	29	0
U3 - Toxicity prior Beta(0.3, 0.7)	11.8	7.7	12.5	44.2	23.8	0
U2 - Toxicity prior Beta(0.5, 0.5)	6.2	4.7	12	45.9	31.2	0
U3 - Toxicity prior Beta(0.5, 0.5)	10.2	7	11.7	45.5	25.6	0
U2 - Toxicity prior Beta(1, 1)	6.5	5.3	10.2	45.1	32.9	0
U3 - Toxicity prior Beta(1, 1)	9.8	6.9	11.8	45.1	26.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.8	6.1	10.8	5.9	
U3 - Toxicity prior Beta(0.15, 0.85)	3.4	4	6.3	11	5.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.7	5.8	11	6.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.4	3.8	5.9	11.2	5.7	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.6	5.5	11.2	6.4	
U3 - Toxicity prior Beta(0.5, 0.5)	3.3	3.7	5.6	11.5	5.8	
U2 - Toxicity prior Beta(1, 1)	3.2	3.6	5.4	11.2	6.5	
U3 - Toxicity prior Beta(1, 1)	3.3	3.7	5.7	11.4	5.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.8	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.8	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.6	1.5	2.7	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.7	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.6	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.6	1.2	
U2 - Toxicity prior Beta(1, 1)	0	2.6	1.5	2.6	1.3	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.5	2.6	1.1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13	17.8	49.2	17.1	2.6	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	17.2	19.8	45.3	15.8	1.7	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	13.6	15.2	47.7	19.9	3.4	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	17.3	19	44.7	16.3	2.5	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	12.3	15.7	44.6	23.1	4.1	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	17.1	18	42.3	19.6	2.8	0.2
U2 - Toxicity prior Beta(1, 1)	11.4	16.6	43.9	23.8	4	0.3
U3 - Toxicity prior Beta(1, 1)	16.5	19.5	42.3	18.9	2.5	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.6	7.1	10	5.7	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	6.1	7.1	9.9	5.5	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	5.3	6.8	10	6	1.9	
U3 - Toxicity prior Beta(0.3, 0.7)	5.8	6.7	10	5.8	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	4.8	6.5	9.7	6.7	2.3	
U3 - Toxicity prior Beta(0.5, 0.5)	5.3	6.5	9.8	6.5	1.9	
U2 - Toxicity prior Beta(1, 1)	4.9	6.5	9.6	6.7	2.3	
U3 - Toxicity prior Beta(1, 1)	5.3	6.4	9.8	6.5	2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.7	2.3	1.9	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.6	2.2	1.8	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.6	2.2	1.9	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.5	2.2	1.8	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	2.2	1.9	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.4	2.1	1.8	0.5	
U2 - Toxicity prior Beta(1, 1)	0.5	3.3	2.1	1.9	0.6	
U3 - Toxicity prior Beta(1, 1)	0.6	3.4	2.1	1.8	0.5	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7.8	44.5	35.2	11.1	1.4	0
U3 - Toxicity prior Beta(0.15, 0.85)	10.7	48.4	32	8.2	0.7	0
U2 - Toxicity prior Beta(0.3, 0.7)	8.1	43	35	12.2	1.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	10.5	48.1	31.6	9	0.8	0
U2 - Toxicity prior Beta(0.5, 0.5)	6.9	41.4	36.5	13.6	1.6	0
U3 - Toxicity prior Beta(0.5, 0.5)	11.5	45.4	31.1	11.2	0.7	0.1
U2 - Toxicity prior Beta(1, 1)	7.2	41.1	34.7	15.5	1.5	0
U3 - Toxicity prior Beta(1, 1)	11.9	46.2	29.4	11.4	1	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	12.9	8.9	3.2	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	13.7	8.3	2.9	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	12.6	8.9	3.5	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	13.6	8.2	3.1	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	12.5	8.9	3.7	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	13.4	8.3	3.4	0.5	
U2 - Toxicity prior Beta(1, 1)	4.2	12.5	8.8	3.8	0.8	
U3 - Toxicity prior Beta(1, 1)	4.4	13.4	8.2	3.4	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	2.1	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	2	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.1	2.2	1.1	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.1	2	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3	2.2	1	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3	2	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	0.2	3	2.2	1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.2	3	2	0.9	0.1	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	24	48.5	12.7	3.9	0.8	10.1
U3 - Toxicity prior Beta(0.15, 0.85)	27.3	48	10.7	3.6	0.4	10
U2 - Toxicity prior Beta(0.3, 0.7)	20.9	50.5	13.8	5	1.2	8.6
U3 - Toxicity prior Beta(0.3, 0.7)	24.1	50.7	10.9	4.5	0.7	9.1
U2 - Toxicity prior Beta(0.5, 0.5)	19.5	50.1	16.8	5.6	1	7
U3 - Toxicity prior Beta(0.5, 0.5)	22.6	50.8	14.5	4.2	0.7	7.2
U2 - Toxicity prior Beta(1, 1)	19	50.1	16.9	5.9	1	7.1
U3 - Toxicity prior Beta(1, 1)	23.5	50.4	13.4	4.7	0.6	7.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.2	11.1	4.9	2.5	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	10.7	11	4.6	2.4	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9.7	11.1	5.1	2.8	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	10.3	10.9	4.8	2.6	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	8.6	11.1	5.8	3	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	9.2	11.1	5.6	2.9	0.8	
U2 - Toxicity prior Beta(1, 1)	8.6	11.2	5.8	3	0.9	
U3 - Toxicity prior Beta(1, 1)	9.2	11.1	5.6	2.9	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	5.1	1.4	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	5	1.3	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	5	1.5	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.9	1.4	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	4.6	1.6	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	4.6	1.6	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.8	4.6	1.6	1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.9	4.6	1.5	1	0.2	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	23.4	58.5	12.2	1.9	0	4
U3 - Toxicity prior Beta(0.15, 0.85)	27.4	55.5	10.9	1.1	0.1	5
U2 - Toxicity prior Beta(0.3, 0.7)	20.8	59.1	15.1	1.4	0.1	3.5
U3 - Toxicity prior Beta(0.3, 0.7)	27.1	56.2	11.4	1.5	0	3.8
U2 - Toxicity prior Beta(0.5, 0.5)	20.2	60.1	14.9	1.3	0.2	3.3
U3 - Toxicity prior Beta(0.5, 0.5)	25.9	56.9	12.7	1	0.1	3.4
U2 - Toxicity prior Beta(1, 1)	20.5	59.3	15.5	1.2	0.2	3.3
U3 - Toxicity prior Beta(1, 1)	26	56.4	12.5	1	0.2	3.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.5	12.5	5.1	2.2	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	10.1	12.3	4.9	2.1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	8.9	12.4	5.4	2.4	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9.7	12.2	5.1	2.3	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	8.1	12.7	5.9	2.6	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	12.4	5.6	2.4	0.6	
U2 - Toxicity prior Beta(1, 1)	8.1	12.5	5.9	2.6	0.7	
U3 - Toxicity prior Beta(1, 1)	8.8	12.3	5.7	2.5	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	5.4	1.5	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	1	5.2	1.5	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	5.2	1.6	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	1	5.2	1.5	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	4.9	1.7	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	4.8	1.6	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	1	4.9	1.7	1	0.2	
U3 - Toxicity prior Beta(1, 1)	1	4.8	1.6	0.9	0.2	